Yared Tsegaye Gizaw

Software Engineer | Data Scientist

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OBJECTIVE

MS in Data Science student with a strong background in Software Engineering and hands-on experience building scalable web, mobile, and AI-powered applications. Skilled in Python, Java, and JavaScript, with expertise in full-stack development, cloud deployment, and system optimization. Passionate about designing efficient, reliable software solutions that combine data-driven insights with real-world impact.

EDUCATION

Fordham University Aug 2025-May 2027 Expected
M.S. in Data Science *New York, United States*

- Relevant Coursework: Data Mining, Maths for Data Science, Algorithms for Data Science, Machine Learning, RL

Addis Ababa University (AAiT) Sep 2019–Jul 2024
Bachelor of Science (B.S.) in Software Engineering *Addis Ababa, Ethiopia*

- GPA: 3.75/4.0
- Relevant Coursework: Data Structures and Algorithms, Machine Learning, AI, Web and Mobile Development, Operating Systems, Cyber Security, Object-Oriented Programming.

A2SV | Africa to Silicon Valley Dec 2021–Nov 2022
Data Structures and Algorithms *Addis Ababa, Ethiopia*

- Relevant Coursework: Data Structures and Algorithms, Algorithm Design and Analysis

RELEVANT WORK EXPERIENCE

Fordham University Aug 2025 – Present
Graduate Research Assistant *New York, United States*

- Working on combinatorial optimization problems with Integer and Mixed Integer Linear Programming
- Exploring the potential application of Reinforcement Learning techniques to improve solutions for Integer Linear Programming problems.

Turing May. 2025 – Aug. 2025
Python Developer, Data Scientist, AI Trainer *Palo Alto, CA, United States*

- Evaluated model performance on Text generation, Code generation, and performing sophisticated Math problems, and more, using Reinforcement Learning from Human Feedback (RLHF) techniques.

BidWiseAI Feb 2025 – Aug 2025
Full Stack Developer / RAG Pipeline *London, UK*

- Created a RAG pipeline to process procurement documents and generate answers using the Gemini API.
- Implemented pixel-perfect user interfaces from Figma designs using React.
- Implemented and deployed FastAPI backend services on Azure Container Apps. Used Docker to package the application into a container image, and then deployed it to a Container Apps environment.

Aladia Jan 2025 – Feb 2025
Frontend QA Developer *Brescia, Italy*

- Writing unit and end-to-end tests using Storybook for a Nuxt.js platform.
- Evaluating the performance, features, and cost of AI automation tools like Testim, Playwright, Katalon for selection.

IPVideo Aug 2024 - Jan 2025
Frontend and Security Engineer *London, UK*

- Created a CCTV camera tracking app using React Native and Strapi.
- Used OpenCV to convert snapshot videos to frames, utilized multiple LLMs like(OpenAI, Claude Anthropic, and Google Vision) to process and detect objects, and send a notification to users for suspicious activity.
- Enhanced the security of the web/mobile application by implementing input validation and XSS attack handling.

Eskalate Feb 2023–June 2023
Mobile Application Developer *San Francisco Bay Area*

- Spearheaded development of the Ride Sharing app using Flutter, Dart, Google Maps API, and Clean Test-Driven Development architecture
- Enhanced A2SV portal serving 3,000 users, and is projected to grow to 5,000 users within the next 2 years.
- Ensured reliability through rigorous testing, achieving 98% code coverage as measured with SonarCloud.

Temari-bet Mar 2022–Jun 2022
Full Stack Developer *Addis Ababa, Ethiopia*

- Developed a website connecting parents with tutors for their children using Next.js, Node.js, and Express.js
- Digitized communication channels from Telegram to an End-to-end chat system.
- Transformed the storage system of some projects from MongoDB Atlas to Amazon S3 and Cloudinary, providing 100% data safety and scalability.

TECHNICAL SKILLS

Programming & Scripting: Python, JavaScript/TypeScript, SQL, NoSQL, C/C++, Rust, Java

Web & Mobile Development: MERN, MEAN, Django, FastAPI, Node.js, Express, React.js, Angular, Vue.js, React Native, Flutter

Data Science: TensorFlow, Scikit-learn, Pandas, ML, Data Mining, Probability & Statistics, Data Science, PyTorch

Tools & DevOps: Docker, Git, CI/CD, Azure DevOps, Linux Systems, Selenium, Hadoop

Additional: Object-oriented programming, Cloud Services (AWS, GCP), Spring-boot

VOLUNTARY ACTIVITIES

AddisCoder 2023 by **Professor Jelani Nelson** Jul 2023–Aug 2023
Teaching Assistant *Addis Ababa, Ethiopia*

- I volunteered as a teaching assistant for AddisCoder 2023, a free program in Addis Ababa introducing high schoolers to Python, data structures, and algorithms, alongside Professor Jelani Nelson from UC Berkeley.

Addis Ababa Institute of Technology University Sep 2023 – Feb 2024
AI Trainer *Addis Ababa, Ethiopia*

- Created a Voice-Based Agricultural Information System for the Amharic language using a pre-trained LLAMA model.
- Performed the Low-Rank Adaptation (LoRA) method for the fine-tuning process with the questions and answers dataset.
- The project was created for Ethiopian farmers to have a voice-based back-and-forth conversation about agriculture with the model in the Amharic language. It was built in collaboration with Addis Ababa University.

ACHIEVEMENTS

- **2nd place in the Ethiopian Collegiate Programming Contest (EtCPC)**, which is a sub-part of **ICPC**) and passed to the **ACPC** held in Egypt.
- Won a team project competition at Eskalate with Atrons and RideShare, earning a prize of 8000$ and 2000$.
- **IELTS overall score of 7.5**
- **Participated in the OBSS Codemaster competition** and demonstrated strong coding skills.
- **Presented on National Television** for creating impactful projects for agriculture in the country using AI and LLM.